「Galois group of $(x^2-2)(x^2-3)$ over $\mathbb{Q}$.

Since the polynomial $(x^2-2)(x^2-3)$ has four distinct roots $\pm\sqrt{2}, \pm\sqrt{3}$, it is separable.
And, since $\sqrt{2} \notin \mathbb{Q}(\sqrt{3})$, $[\mathbb{Q}(\sqrt{2},\sqrt{3}):\mathbb{Q}] = [\mathbb{Q}(\sqrt{2},\sqrt{3}), \mathbb{Q}(\sqrt{3})]\cdot[\mathbb{Q}(\sqrt{3}):\mathbb{Q}] = 4$.
So, $|\mathrm{Aut}(K/\mathbb{Q})| = [K:\mathbb{Q}] = 4$ where $K = \mathbb{Q}(\sqrt{2},\sqrt{3})$, the splitting field over $\mathbb{Q}$.
Since the automorphism leaving $\mathbb{Q}$ fixed must be conjugate the roots,
there are four possible automorphisms;

$$\begin{cases}\sqrt{2}\mapsto\sqrt{2}\\ \sqrt{3}\mapsto\sqrt{3}\end{cases},\quad \begin{cases}\sqrt{2}\mapsto-\sqrt{2}\\ \sqrt{3}\mapsto\sqrt{3}\end{cases},\quad \begin{cases}\sqrt{2}\mapsto\sqrt{2}\\ \sqrt{3}\mapsto-\sqrt{3}\end{cases},\quad \begin{cases}\sqrt{2}\mapsto-\sqrt{2}\\ \sqrt{3}\mapsto-\sqrt{3}\end{cases}.$$

Let $\sigma : K \to K ; a + b\sqrt{2} + c\sqrt{3} + d\sqrt{6} \mapsto a - b\sqrt{2} + c\sqrt{3} - d\sqrt{6}$

$\tau : K \to K ; a + b\sqrt{2} + c\sqrt{3} + d\sqrt{6} \mapsto a + b\sqrt{2} - c\sqrt{3} - d\sqrt{6}$.

(That is, $\sigma : \begin{cases}\sqrt{2}\mapsto-\sqrt{2}\\ \sqrt{3}\mapsto\sqrt{3}\end{cases}$ and $\tau : \begin{cases}\sqrt{2}\mapsto\sqrt{2}\\ \sqrt{3}\mapsto-\sqrt{3}\end{cases}$, and we can observe that

$\sigma\tau : \begin{cases}\sqrt{2}\mapsto-\sqrt{2}\\ \sqrt{3}\mapsto-\sqrt{3}\end{cases}$, $\tau\sigma : \begin{cases}\sqrt{2}\mapsto-\sqrt{2}\\ \sqrt{3}\mapsto-\sqrt{3}\end{cases}$, so $\sigma^2 = \tau^2 = 1$ and $\sigma\tau = \tau\sigma$

Now, $\mathrm{Gal}(K/\mathbb{Q}) = \{1, \sigma, \tau, \sigma\tau\} \cong \mathbb{Z}_2 \times \mathbb{Z}_2$, corresponding σ to $(1,0)$ and τ to $(0,1)$.
Moreover,

$$\begin{tikzcd}
 & \mathbb{Q}(\sqrt{2},\sqrt{3}) = K^{\{id\}} \arrow[dl, dash] \arrow[d, dash] \arrow[dr, dash] & \\
\mathbb{Q}(\sqrt{2}) = K^{\langle\tau\rangle} \arrow[dr, dash] & \mathbb{Q}(\sqrt{6}) = K^{\langle\sigma\tau\rangle} \arrow[d, dash] & \mathbb{Q}(\sqrt{3}) = K^{\langle\sigma\rangle} \arrow[dl, dash] \\
 & \mathbb{Q} = K^{\mathrm{Gal}(K/\mathbb{Q})} &
\end{tikzcd}$$

Note; $\sqrt{2} \in \mathbb{Q}(\sqrt{3})$ because; if $\sqrt{2} = a + b\sqrt{3}$ for some $a, b \in \mathbb{Q}$, then $2 = a^2 + 3b^2 + 2ab\sqrt{3}$.
it is contradiction

「Galois group of x^3-2 over $\mathbb{Q}$.

Denote $\zeta_3 = e^{i\cdot\frac{2\pi}{3}} = \frac{1}{2}(-1+i\cdot\sqrt{3})$, the 3th primitive root of unity.

Since $\zeta_3^3=1$, the set $\{\sqrt[3]{2}, \sqrt[3]{2}\,\zeta_3, \sqrt[3]{2}\cdot\zeta_3^2\}$ are all roots of x^3-2 over $\mathbb{Q}$.

So, the splitting field is $K=\mathbb{Q}(\sqrt[3]{2},\zeta_3). = \mathbb{Q}(\sqrt[3]{2}, i\sqrt{3})$.

observe that: the minimal polynomial of $\sqrt[3]{2}$ over $\mathbb{Q}$ is x^3-2, and

" ζ_3 over $\mathbb{Q}$ is x^2+x+1.

And, since x^2+x+1 is irreducible over $\mathbb{Q}(\sqrt[3]{2})$, (it cannot have root in $\mathbb{Q}(\sqrt[3]{2})$)

$$[\mathbb{Q}(\sqrt[3]{2},\zeta_3):\mathbb{Q}] = [\mathbb{Q}(\sqrt[3]{2},\zeta_3):\mathbb{Q}(\sqrt[3]{2})]\cdot[\mathbb{Q}(\sqrt[3]{2}):\mathbb{Q}] = 2\cdot 3 = 6.$$

the tower Lemma gave. And, since x^3-2 is separable, $|Aut(K/\mathbb{Q})| = [K:\mathbb{Q}] = 6$.

Now, Construct explicity the $Aut(K/\mathbb{Q})$.

Since any automorphism of K leaving $\mathbb{Q}$ fixed must Conjugate the roots,

it must maps $\sqrt[3]{2}$ to $\sqrt[3]{2}$ or $\sqrt[3]{2}\,\zeta_3$ or $\sqrt[3]{2}\cdot\zeta_3^2$

and maps ζ_3 to ζ_3 or ζ_3^2.

So, there are exactly 6 possiblities. Consider two automorphisms

σ satisfying $\sigma(\sqrt[3]{2}) = \sqrt[3]{2}\cdot\zeta_3$ and $\sigma(\zeta_3)=\zeta_3$

and τ satisfying $\tau(\sqrt[3]{2}) = \sqrt[3]{2}$ and $\tau(\zeta_3) = \zeta_3^2 = -1-\zeta_3$.

Then, $\sigma^3: \begin{cases} \sqrt[3]{2} \mapsto \sqrt[3]{2}\,\zeta_3 \mapsto \sqrt[3]{2}\cdot\zeta_3^2 \mapsto \sqrt[3]{2}\,\zeta_3^3 = \sqrt[3]{2} \\ \zeta_3 \mapsto \zeta_3. \end{cases}$, $\tau^2: \begin{cases} \sqrt[3]{2} \mapsto \sqrt[3]{2} \\ \zeta_3 \mapsto \zeta_3^2 \mapsto \zeta_3^4 = 1-\zeta_3 = \zeta_3 \end{cases}$

are actually identities. Moreover,

$\sigma^2: \begin{cases} \sqrt[3]{2} \mapsto \sqrt[3]{2}\cdot\zeta_3^2 \\ \zeta_3 \mapsto \zeta_3 \end{cases}$, $\sigma\tau: \begin{cases} \sqrt[3]{2} \mapsto \sqrt[3]{2}\cdot\zeta_3 \\ \zeta_3 \mapsto \zeta_3^2 \end{cases}$, $\sigma^2\tau: \begin{cases} \sqrt[3]{2} \mapsto \sqrt[3]{2}\,\zeta_3^2 \\ \zeta_3 \mapsto \zeta_3^2 \end{cases}$, $\tau\sigma: \begin{cases} \sqrt[3]{2} \mapsto \sqrt[3]{2}\cdot\zeta_3^2 \\ \zeta_3 \mapsto \zeta_3^2 \end{cases}$

So, $Aut(K/\mathbb{Q}) = \{1,\sigma,\sigma^2,\tau,\sigma\tau,\sigma^2\tau\} \cong S_3$, Corresponding σ to $(1\,2\,3)$, τ to $(1\,2)$

Moreover,

$$\begin{tikzcd}
 & & \mathbb{Q}(\sqrt[3]{2},\zeta_3) = K^{\{id\}} \arrow[dll, dash] \arrow[d, dash] \arrow[dr, dash] \arrow[drr, dash] & & \\
\mathbb{Q}(\zeta_3) = K^{\langle\sigma\rangle} \arrow[drr, dash] & & \mathbb{Q}(\sqrt[3]{2}) = K^{\langle\tau\rangle} \arrow[d, dash] & \mathbb{Q}(\sqrt[3]{2}\cdot\zeta_3^2) = K^{\langle\sigma\tau\rangle} \arrow[dl, dash] & \mathbb{Q}(\sqrt[3]{2}\cdot\zeta_3) = K^{\langle\sigma^2\tau\rangle} \arrow[dll, dash] \\
 & & \mathbb{Q} = K^{\langle\sigma,\tau\rangle} & &
\end{tikzcd}$$

How to find a fixed field corresponding to $\langle\sigma\tau\rangle$?

As we know, the $Tr_{K/K^{\langle\sigma\tau\rangle}}$ is invariant under any automorphism in $\langle\sigma\tau\rangle$.

So, Calculate:

$$Tr(\sqrt[3]{2}) = id(\sqrt[3]{2}) + (\sigma\circ\tau)(\sqrt[3]{2}) = \sqrt[3]{2} + \sqrt[3]{2}\,\zeta_3 = \sqrt[3]{2}(1+\zeta_3) = \sqrt[3]{2}\cdot(-\zeta_3^2).$$

Therefore, $\sqrt[3]{2}\,\zeta_3^2$ is fixed by id and $\sigma\circ\tau$, so $\mathbb{Q}(\sqrt[3]{2}\cdot\zeta_3^2) \subseteq K^{\langle\sigma\tau\rangle}$.

Moreover, $[\mathbb{Q}(\sqrt[3]{2},\zeta_3) : \mathbb{Q}(\sqrt[3]{2}\cdot\zeta_3^2)] = 2$, so the Galois theorem gives the result.

「Galois group of x^8-2 over $\mathbb{Q}$.

「Galois group of x^5-3 over $\mathbb{Q}(\zeta)$, where ζ is a primitive fifth root of unity.

Observe that: x^5-3 has five distinct roots, $\sqrt[5]{3}\cdot\zeta^k$ $(k=0,1,\cdots,4)$.
Since the minimal polynomial of ζ over $\mathbb{Q}$ is $x^4+x^3+x^2+x+1$,
therefore $[\mathbb{Q}(\zeta):\mathbb{Q}]=4$. And, the minimal polynomial of $\sqrt[5]{3}$ over $\mathbb{Q}$ is x^5-3,
therefore $[\mathbb{Q}(\sqrt[5]{3}):\mathbb{Q}]=5$. Therefore, the splitting field of x^5-3 over $\mathbb{Q}$,

$$K=\mathbb{Q}(\sqrt[5]{3},\ \sqrt[5]{3}\zeta^1,\ \sqrt[5]{3}\zeta^2,\ \sqrt[5]{3}\zeta^3,\ \sqrt[5]{3}\zeta^4)=\mathbb{Q}(\sqrt[5]{3},\zeta)=\mathbb{Q}(\sqrt[5]{3})\mathbb{Q}(\zeta)$$

has degree of 20 over $\mathbb{Q}$, that is, $[K:\mathbb{Q}]=20$.
More precisely, by the tower lemma, $[K:\mathbb{Q}]$ is divided by $[\mathbb{Q}(\sqrt[5]{3}):\mathbb{Q}]$ and $[\mathbb{Q}(\zeta):\mathbb{Q}]$.
Since 5 and 4 are relatively prime, $[K:\mathbb{Q}]\geq 20$.
And, $x^5-3\in\mathbb{Q}(\zeta)[x]$ has $\sqrt[5]{3}$ as a root, so

$$[\mathbb{Q}(\sqrt[5]{3},\zeta):\mathbb{Q}(\zeta)]\cdot[\mathbb{Q}(\zeta):\mathbb{Q}]\leq 20.$$

Consequently, $[K:\mathbb{Q}]=20$. Now, by the Fundamental Theorem of Galois Theory,
$K/\mathbb{Q}(\zeta)$ is also Galois, with $[K:\mathbb{Q}(\zeta)]=5$.
Therefore, $\mathrm{Gal}(K/\mathbb{Q}(\zeta))$ has order of 5, so it must be a cyclic group of order 5. Consequently, the automorphism

$$\sigma:\begin{cases}\sqrt[5]{3}\mapsto\sqrt[5]{3}\cdot\zeta\\ \zeta\mapsto\zeta\end{cases}$$ (It is given by the conjugate isomorphism theorem).

has order of 5, therefore $\mathrm{Gal}(K/\mathbb{Q}(\zeta))=\langle\sigma\rangle\cong\mathbb{Z}_5$. ☑

Latticeally,

$$\begin{tikzcd}
 & \mathbb{Q}(\sqrt[5]{3},\zeta)=K^{\{id\}} \arrow[dl, dash, "5"'] \arrow[dr, dash, "4"] & \\
\mathbb{Q}(\zeta)=K^{\langle\sigma\rangle} \arrow[dr, dash, "4"'] & & \mathbb{Q}(\sqrt[5]{3})=K^{\langle\tau\rangle} \arrow[dl, dash, "5"] \\
 & \mathbb{Q}=K^{\mathrm{Aut}(K/\mathbb{Q})=\langle\sigma,\tau\rangle} &
\end{tikzcd}$$

where $\tau:\begin{cases}\sqrt[5]{3}\mapsto\sqrt[5]{3}\\ \zeta\mapsto\zeta^2.\end{cases}$ Therefore, $\mathrm{Aut}(K/\mathbb{Q})\cong\langle(1\ 2\ 3\ 4\ 5),(2\ 3\ 5\ 4)\rangle$.

[Galois group of x^6+3 over $F=\mathbb{Q}$ or $\mathbb{F}_5$ or $\mathbb{F}_7$. (UCLA Qual, 2022 Fall)
For each case, determine the splitting field K and evaluate $[K:F]$.
And, find the Galois group $\mathrm{Gal}(K/F)$.
(Note: the field of characteristic 0 or finite field are perfect.).

1) $F=\mathbb{Q}$.
By the eisenstein creterion, all coefficient of x^6+3 except the leading coefficients are contained in $3\mathbb{Z}$, but the square of constant term $3\notin 3^2\mathbb{Z}$, the given polynomial x^6+3 is irreducible over $\mathbb{Q}$. The x^6+3 has 6 roots,

$$\sqrt[6]{3}\cdot w^{2k+1}\ (k=0,1,\cdots,5) \text{ where } w=\exp\left(\frac{i\pi}{6}\right)=\frac{\sqrt{3}}{2}+i\cdot\frac{1}{2}.$$

Therefore, the splitting field is $K=\mathbb{Q}(\sqrt[6]{3}\cdot w,\ \sqrt[6]{3}\cdot w^3,\ \cdots,\ \sqrt[6]{3}\cdot w^{11})$,
Clearly $\mathbb{Q}(\sqrt[6]{3}\cdot w)\subseteq K$, by the definition of generated field.
Meanwhile, $(\sqrt[6]{3}\cdot w)^3=\sqrt{3}\cdot w^3=\sqrt{3}\cdot\exp(i\pi/2)=\sqrt{3}\cdot i$ and $w^2=\frac{1}{2}+\frac{i\sqrt{3}}{2}$ implies $w^2\in\mathbb{Q}(\sqrt[6]{3}\cdot w)$, so $K=\mathbb{Q}(\sqrt[6]{3}\cdot w)$. Finally, since x^6+3 is irreducible, $[K:\mathbb{Q}]=6$.
Moreover, $\mathrm{Gal}(K/\mathbb{Q})\cong S_3$, because: define two automorphisms on K as:
let $\theta=\sqrt[6]{3}\cdot w$ and $\lambda=w^2$. Then, $\theta^3=i\sqrt{3}$ and $\lambda=\frac{1}{2}+\frac{\theta^3}{2}$. Now, let

$$\sigma:\theta\mapsto\theta\cdot\lambda \quad\text{and}\quad \tau:\theta\mapsto\theta\lambda^2.$$

Then, $\sigma(\lambda)=\sigma(\frac{1}{2}+\frac{1}{2}\theta^3)=\frac{1}{2}+\frac{1}{2}\cdot\theta^3\cdot\lambda^3=\frac{1}{2}+\frac{1}{2}\cdot i\sqrt{3}\cdot w^6=\frac{1}{2}(1-i\sqrt{3})=w^{10}=\lambda^{-1}$.
So $\sigma^2(\theta)=\sigma(\theta\lambda)=\theta\lambda\lambda^{-1}=\theta$, it means $\sigma^2=id$.
And $\tau(\lambda)=\frac{1}{2}+\frac{1}{2}\theta^3\cdot\lambda^6=\frac{1}{2}+\frac{1}{2}i\sqrt{3}=\lambda$, so
$\tau^2(\theta)=\tau(\theta\lambda^2)=\theta\lambda^4$, $\tau^3(\theta)=\theta\lambda^6=\theta$, so $\tau^3=id$.
And, $\sigma\tau(\theta)=\sigma(\theta\lambda^2)=\theta\cdot\lambda^{-1}=\theta\cdot\lambda^2$, $(\sigma\tau)^2(\theta)=\sigma\tau(\theta\lambda^2)=\theta\lambda^{-3}=\theta$,
therefore $\sigma\tau\sigma\tau=id$, or $\sigma\tau\sigma=\tau^{-1}$. Now, $\mathrm{Gal}(K/F)=\langle\sigma,\tau\rangle\cong D_6\cong S_3$.

Alternatively, if $\mathrm{Gal}(K/\mathbb{Q})\cong\mathbb{Z}/6\mathbb{Z}$, the abelian group, then every subgroup is normal.
Therefore every intermidate field E satisfies E/F is Galois. (In other say, normal)
Since $\sqrt[6]{3}\cdot w^{11}\cdot\sqrt[6]{3}\cdot w=\sqrt[3]{3}\in K$, the field $\mathbb{Q}(\sqrt[3]{3})$ is the intermidate field,
but as we know, the minimal polynomial of $\sqrt[3]{3}$ is x^3-3 over $\mathbb{Q}$, but $\mathbb{Q}(\sqrt[3]{3})$ does not have other roots of x^3-3. So, it is contradiction.
By the knowledge in the group theory, there are only two group of order 6, S_3 and $\mathbb{Z}/6\mathbb{Z}$, so $\mathrm{Gal}(K/\mathbb{Q})\cong S_3$. ☒

「Galois group of x^6+3 over $F = \mathbb{Q}$ or $\mathbb{F}_5$ or $\mathbb{F}_7$. (UCLA Qual, 2022 Fall)
For each case, determine the splitting field K and evaluate $[K:F]$.
And, find the Galois group $Gal(K/F)$.
(Note: the field of characteristic 0 or finite field are perfect.)」.

2). $F = \mathbb{F}_5$

「Galois group of x^6+3 over $F = \mathbb{Q}$ or $\mathbb{F}_5$ or $\mathbb{F}_7$. (UCLA Qual, 2022 Fall)
For each case, determine the splitting field K and evaluate $[K:F]$.
And, find the Galois group $Gal(K/F)$.
(Note: the field of characteristic 0 or finite field are perfect.).

3). $F = \mathbb{F}_7$